Size distributions: what N, $N(D)$, $n(D)$ and $\mathrm{d}N/\mathrm{d}D$ mean, and what to fit

1. The quantities. A probe counts particles in a sampled volume and sorts them into size bins i with lower and upper edges D_i^-, D_i^+, width $\Delta D_i = D_i^+ - D_i^-$ (not constant across bins) and a representative size D_i (the bin midpoint).

Symbol	Name	Units	Meaning
N_T	total number concentration	cm^{-3}	all particles per unit volume of air
ΔN_i	bin concentration	cm^{-3}	particles per unit volume in bin i
$n(D) \equiv \dfrac{\mathrm{d}N}{\mathrm{d}D}$	number distribution function	$\mathrm{cm}^{-3}\,\mu\mathrm{m}^{-1}$	particles per unit volume *per unit diameter*
$f(D)$	probability density of size	$\mu\mathrm{m}^{-1}$	$n(D)/N_T$; integrates to 1

The measured estimate of $n(D)$ in bin i is the bin concentration divided by the bin width,

$$n(D_i) \approx \frac{\Delta N_i}{\Delta D_i} \quad \Longleftrightarrow \quad \Delta N_i \approx n(D_i)\,\Delta D_i, \tag{1}$$

which is why a code that stores `dNdD` recovers the per-bin concentration as `dNdD * bin_width`. The totals follow as a sum over bins or an integral over size:

$$N_T \;=\; \sum_i \Delta N_i \;=\; \int_0^\infty n(D)\,\mathrm{d}D. \tag{2}$$

Integrating ΔN_i (cm^{-3}) over D does *not* give N_T; only $n(D)$ is integrated.

2. The gamma form (Johnson & Kliche 2020, *Atmosphere* 11, 333, Eqs. 1–2). The distribution is written as total concentration times a probability density:

$$n(D) \;=\; N_T\, f(D;\mu,\lambda), \qquad f(D;\mu,\lambda) \;=\; \frac{\lambda^{\mu+1}}{\Gamma(\mu+1)}\, D^\mu e^{-\lambda D}, \quad D>0, \tag{3}$$

with $\int_0^\infty f\,\mathrm{d}D = 1$. Multiplying out gives the familiar three-parameter form

$$n(D) \;=\; N_0\, D^\mu e^{-\lambda D}, \qquad N_0 \;=\; \frac{N_T\,\lambda^{\mu+1}}{\Gamma(\mu+1)}, \qquad N_T \;=\; \int_0^\infty N_0 D^\mu e^{-\lambda D}\,\mathrm{d}D \;=\; \frac{N_0\,\Gamma(\mu+1)}{\lambda^{\mu+1}}. \tag{4}$$

- The left side, often written $N(D)$, is the number distribution function $n(D) = \mathrm{d}N/\mathrm{d}D$ ($\mathrm{cm}^{-3}\,\mu\mathrm{m}^{-1}$), *not* a per-bin concentration.
- N_0 is the *intercept* parameter, not the total concentration; its units, $\mathrm{cm}^{-3}\,\mu\mathrm{m}^{-1-\mu}$, depend on μ. It equals N_T only in the special case $\lambda^{\mu+1} = \Gamma(\mu+1)$.
- μ is the shape parameter (dimensionless); λ is the *rate* (slope) parameter ($\mu\mathrm{m}^{-1}$), the inverse of the scale parameter $1/\lambda$.

3. Size distribution versus probability distribution. $n(D)$ is a concentration density: it integrates to N_T, a physical number per volume. Dividing by N_T gives $f(D)$, a probability density of particle size that integrates to 1. So a gamma size distribution is a gamma probability density scaled by N_T, exactly as Eq. (3) writes it. By contrast $n(D)/N_0 = D^\mu e^{-\lambda D}$ integrates to $\Gamma(\mu+1)/\lambda^{\mu+1}$, which is generally not 1.

4. What to fit. $x = D_i$ (bin midpoint) and $y = \mathrm{d}N/\mathrm{d}D$ at D_i, because the model $N_0 D^\mu e^{-\lambda D}$ *is* a density in D. Fitting the per-bin concentration ΔN_i instead would make wide bins look more populated than narrow ones purely because they are wide, and the shape of the fitted curve would follow the bin layout rather than the cloud. Equivalently, one may fit ΔN_i against the model integrated over each bin, $\int_{D_i^-}^{D_i^+} N_0 D^\mu e^{-\lambda D}\,\mathrm{d}D$, which is the more exact treatment for wide bins.

Notation in the literature is not uniform: some authors write $N(D)$ for what is called $n(D)$ here. Read the units to tell which is meant: $\mathrm{cm}^{-3}\,\mu\mathrm{m}^{-1}$ is the distribution function, cm^{-3} is a bin concentration or a total.